Name _____ abc123_____

Homework #2

1. Write the Verilog code for a 4:1 MUX

```
module mux_4to1_case ( input [3:0] a,              // 4-bit input called a
                       input [3:0] b,              // 4-bit input called b
                       input [3:0] c,              // 4-bit input called c
                       input [3:0] d,              // 4-bit input called d
                       input [1:0] sel,            // input sel used to select between a,b,c,d
                       output reg [3:0] out);      // 4-bit output based on input sel

  // This always block gets executed whenever a/b/c/d/sel changes value
  // When that happens, based on value in sel, output is assigned to either a/b/c/d
  always @ (a or b or c or d or sel) begin
    case (sel)
      2'b00 : out = a;
      2'b01 : out = b;
      2'b10 : out = c;
      2'b11 : out = d;
    endcase
  end
endmodule
```

2. There are a number of delays in Verilog. The two we will use most in this course are below. Explain the difference between these two delays:
    - Inertial Delay: Rejects pulses shorter than propagation delay.
    - Transport Delay: Transport does not reject any pulses

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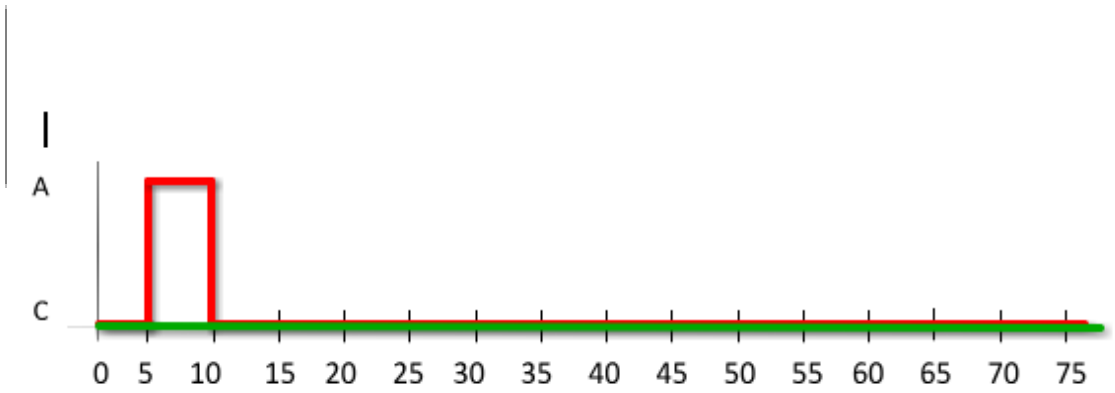
3. You apply a 5ns A=1 pulse to signal A where B=1. Draw the resultant timing diagram for A and C where C is driven by this code:

assign #10 C = A && B

A

C



4. Write Verilog code for a 6 bit adder using a full 1 bit adder modules

```
module fulladd (  input [3:0] a,
                  input [3:0] b,
                  input c_in,
                  output reg c_out,
                  output reg [3:0] sum);

  always @ (a or b or c_in) begin
   {c_out, sum} = a + b + c_in;
 end
endmodule
```